7D_2 is the Case where the curled
geometry is $S^2 \times S^1$, the product of a 2-sphere and a circle.

Notation and Setup:

Starting Metric for this Case: $g_{00}=\frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}$, $g_{11}=-a[t]^2\,B[r]$, $g_{22}=-r^2\,a[t]^2$, $g_{33}=-r^2\,a[t]^2\,\text{Sin}[\theta]^2$, $g_{44}=-a[t]^2\,p_4^2\,\mathcal{D}_4[r]$, $g_{55}=-a[t]^2\,\text{Sin}[\psi_4]^2\,p_4^2\,\mathcal{D}_4[r]$, $g_{66}=-a[t]^2\,p_6^2\,\mathcal{D}_6[r]$, $a(t)$ not constant

Corresponding Line Element

$$ds^2 = \frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}\,dt^2 - a[t]^2\,B[r]\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2 - a[t]^2\,p_4^2\,\mathcal{D}_4[r]\,d\psi_4^2 - a[t]^2\,\text{Sin}[\psi_4]^2\,p_4^2\,\mathcal{D}_4[r]\,d\psi_5^2 - a[t]^2\,p_6^2\,\mathcal{D}_6[r]\,d\psi_6^2$$

Output Display Simplification - Parameters are sometimes suppressed
Prime means derivative wrt r or ψ.
Dot means derivative wrt t.
Examples: **$A = A(t,r)$**, **$A' = \partial_r A(t,r)$** and **$\dot{A} = \partial_t A(t,r)$**

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameter a, (2) $C = 1$, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the $a(t)$ formula).

FG-GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameters a and C, and (2) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation.

a: GRW Equations

The starting metric and line element above are the result of the following 4-space and curled space assumptions.

- 4-space:

  $A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6\,k+\Lambda\,a[t]^2}$, the projected Free-gauge GRW formula for 7D

  $B(r)$, a free parameter

  $C(r) = 1$, the standard GRW value

- Curled-space:

  The line element is built from $S^2 \times S^1$, product of a compact 2-sphere of radius p_4 with a circle of radius p_6:

  S^2 metric:

  (1) $ds^2 = p_4^2 [\, d\psi_4^2 + \sin^2\psi_4 \, d\psi_5^2 \,]$

  S^1 metric

  (2) $ds^2 = p_6^2 \, d\psi_6^2$

  After applying the scale factor $a(t)$ and free parameters $\mathbb{D}_4$ and $\mathbb{D}_6$, the transverse curled line element is

  (3) $ds^2 = a[t]^2 \{ \mathbb{D}_4[r] p_4^2 [\, d\psi_4^2 + \sin^2\psi_4 \, d\psi_5^2 \,] + \mathbb{D}_6[r] p_6^2 \, d\psi_6^2 \}$

  where we allow $\mathbb{D}_4$ and $\mathbb{D}_6$ to be functions of r.

  Thus, our curled space assumptions are:

  - $\mathbb{D}_4(r)$ is a free parameter
  - $\mathbb{D}_6(r)$ is a free parameter
  - Radii p_4 and p_6 are free constants

  As an aside, notice that there is no $\mathbb{D}_5$ parameter.

The initial field equations have 3 unknowns, $\mathbb{B}$, $\mathbb{D}_4$, and $\mathbb{D}_6$.

After subtracting Λ from both sides, they can be expressed:

R_{00}: $0 = 0$

$$R_{11}: \quad -\frac{5k}{a^2} + \frac{\mathbb{B}'}{a^2 r \mathbb{B}^2} + \frac{\mathbb{B}' \mathbb{D}_4'}{2 a^2 \mathbb{B}^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{2 a^2 \mathbb{B} \mathbb{D}_4^2} + \frac{\mathbb{B}' \mathbb{D}_6'}{4 a^2 \mathbb{B}^2 \mathbb{D}_6} + \frac{(\mathbb{D}_6')^2}{4 a^2 \mathbb{B} \mathbb{D}_6^2} - \frac{\mathbb{D}_4''}{a^2 \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{D}_6''}{2 a^2 \mathbb{B} \mathbb{D}_6} = 0$$

$$R_{22}: \quad -\frac{5k}{a^2} + \frac{1}{a^2 r^2} - \frac{1}{a^2 r^2 \mathbb{B}} + \frac{\mathbb{B}'}{2 a^2 r \mathbb{B}^2} - \frac{\mathbb{D}_4'}{a^2 r \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{D}_6'}{2 a^2 r \mathbb{B} \mathbb{D}_6} = 0$$

$$R_{33}: \quad -\frac{5k}{a^2} + \frac{1}{a^2 r^2} - \frac{1}{a^2 r^2 \mathbb{B}} + \frac{\mathbb{B}'}{2 a^2 r \mathbb{B}^2} - \frac{\mathbb{D}_4'}{a^2 r \mathbb{B} \mathbb{D}_4} - \frac{\mathbb{D}_6'}{2 a^2 r \mathbb{B} \mathbb{D}_6} = 0$$

$$R_{44}: \quad -\frac{5k}{a^2} + \frac{1}{a^2 p_4^2 \mathbb{D}_4} - \frac{\mathbb{D}_4'}{a^2 r \mathbb{B} \mathbb{D}_4} + \frac{\mathbb{B}' \mathbb{D}_4'}{4 a^2 \mathbb{B}^2 \mathbb{D}_4} - \frac{\mathbb{D}_4' \mathbb{D}_6'}{4 a^2 \mathbb{B} \mathbb{D}_4 \mathbb{D}_6} - \frac{\mathbb{D}_4''}{2 a^2 \mathbb{B} \mathbb{D}_4} = 0$$

$$R_{55}: \quad -\frac{5k}{a^2} + \frac{1}{a^2 p_4^2 \mathbb{D}_4} - \frac{\mathbb{D}_4'}{a^2 r \mathbb{B} \mathbb{D}_4} + \frac{\mathbb{B}' \mathbb{D}_4'}{4 a^2 \mathbb{B}^2 \mathbb{D}_4} - \frac{\mathbb{D}_4' \mathbb{D}_6'}{4 a^2 \mathbb{B} \mathbb{D}_4 \mathbb{D}_6} - \frac{\mathbb{D}_4''}{2 a^2 \mathbb{B} \mathbb{D}_4} = 0$$

$$R_{66}: \quad -\frac{5k}{a^2} - \frac{\mathbb{D}_6'}{a^2 r \mathbb{B} \mathbb{D}_6} + \frac{\mathbb{B}' \mathbb{D}_6'}{4 a^2 \mathbb{B}^2 \mathbb{D}_6} - \frac{\mathbb{D}_4' \mathbb{D}_6'}{2 a^2 \mathbb{B} \mathbb{D}_4 \mathbb{D}_6} + \frac{(\mathbb{D}_6')^2}{4 a^2 \mathbb{B} \mathbb{D}_6^2} - \frac{\mathbb{D}_6''}{2 a^2 \mathbb{B} \mathbb{D}_6} = 0$$

R_{33} and R_{55} are redundant, the same as R_{22} and R_{44}, respectively.

Because we have set $A(t,r) = \mathcal{A}(t) = \frac{6\,a'[t]^2}{-6k + \Lambda\, a[t]^2}$,

(1) R_{00} is an identity $(\Lambda{=}\Lambda)$

(2) Off-diagonal R_{01} (not shown) is an identity $(0{=}0)$.

However, the remaining equations are too difficult to solve without further assumptions.

In 7D_2 we have two distinct curled objects, S^2 and S^1. So we can assume $B = \frac{1}{\mathcal{D}_4}$ or $B = \frac{1}{\mathcal{D}_6}$.

Assumption 1: $\mathcal{D}_4 = \frac{1}{B}$ (inverse for the 2-sphere)

We allow, but do not require, $\mathcal{D}_6 = \mathcal{D}_4$. That is, we make no assumption about this.

Ignoring R_{00}, R_{33}, and R_{55}, the remaining Lambda field equations are now:

$$R_{11}: \quad -\frac{5k}{a^2} + \frac{B'}{a^2 r B^2} + \frac{B'\mathcal{D}_6'}{4a^2 B^2 \mathcal{D}_6} + \frac{(\mathcal{D}_6')^2}{4a^2 B \mathcal{D}_6^2} - \frac{\frac{2(B')^2}{B^3} - \frac{B''}{B^2}}{a^2} - \frac{\mathcal{D}_6''}{2a^2 B \mathcal{D}_6} = 0$$

$$R_{22}: \quad -\frac{5k}{a^2} + \frac{1}{a^2 r^2} - \frac{1}{a^2 r^2 B} + \frac{3B'}{2a^2 r B^2} - \frac{\mathcal{D}_6'}{2a^2 r B \mathcal{D}_6} = 0$$

$$R_{44}: \quad -\frac{5k}{a^2} + \frac{B}{a^2 p_4^2} + \frac{B'}{a^2 r B^2} - \frac{(B')^2}{4a^2 B^3} + \frac{B'\mathcal{D}_6'}{4a^2 B^2 \mathcal{D}_6} - \frac{\frac{2(B')^2}{B^3} - \frac{B''}{B^2}}{2a^2} = 0$$

$$R_{66}: \quad -\frac{5k}{a^2} - \frac{\mathcal{D}_6'}{a^2 r B \mathcal{D}_6} + \frac{3B'\mathcal{D}_6'}{4a^2 B^2 \mathcal{D}_6} + \frac{(\mathcal{D}_6')^2}{4a^2 B \mathcal{D}_6^2} - \frac{\mathcal{D}_6''}{2a^2 B \mathcal{D}_6} = 0$$

R_{22} can be solved for either B or $\mathcal{D}_6$. The latter is a simpler expression:

$\mathcal{D}_6(r) = e^{\int_1^r \frac{-2B+2B^2-10kB^2\rho^2+3\rho B'}{B\rho}\,d\rho}$ where a constant of integration has been rolled into p_6.

This leaves three equations in the single unknown, B:

$$R_{11}: \quad -\frac{10k}{a^2} + \frac{3}{a^2 r^2} - \frac{2}{a^2 r^2 B} + \frac{10kB}{a^2} - \frac{B}{a^2 r^2} - \frac{25k^2 r^2 B}{a^2} + \frac{7B'}{2a^2 r B^2} - \frac{7B'}{2a^2 r B} + \frac{35kr B'}{2a^2 B} - \frac{2(B')^2}{a^2 B^3} - \frac{B''}{2a^2 B^2} = 0$$

$$R_{44}: \quad -\frac{5k}{a^2} + \frac{B}{a^2 p_4^2} + \frac{B'}{2a^2 r B^2} + \frac{B'}{2a^2 r B} - \frac{5kr B'}{2a^2 B} - \frac{(B')^2}{2a^2 B^3} + \frac{B''}{2a^2 B^2} = 0$$

$$R_{66}: \quad \frac{1}{a^2 r^2} + \frac{10kB}{a^2} - \frac{B}{a^2 r^2} - \frac{25k^2 r^2 B}{a^2} - \frac{3B'}{2a^2 r B^2} - \frac{5B'}{2a^2 r B} + \frac{25kr B'}{2a^2 B} + \frac{3(B')^2}{2a^2 B^3} - \frac{3B''}{2a^2 B^2} = 0$$

$3R_{44}+R_{66}$ can be solved for B:

$3R_{44}+R_{66}$:

$$-\frac{15\,k}{a[t]^2}+\frac{1}{r^2\,a[t]^2}+\frac{10\,k\,B[r]}{a[t]^2}-\frac{B[r]}{r^2\,a[t]^2}-\frac{25\,k^2\,r^2\,B[r]}{a[t]^2}+\frac{3\,B[r]}{a[t]^2\,p_4^2}-\frac{B'[r]}{r\,a[t]^2\,B[r]}+\frac{5\,k\,r\,B'[r]}{a[t]^2\,B[r]}=0$$

$$B(r)=-\frac{3\left(-r\,p_4^2+5\,k\,r^3\,p_4^2\right)}{-3\,r^3+3\,r\,p_4^2-10\,k\,r^3\,p_4^2+15\,k^2\,r^5\,p_4^2+3\,\zeta\,p_4^2}$$

where a ζ is a constant of integration.

The remaining three equations can be simplified by being put over common denominators and then setting the numerators to zero:

R_{11}: $9\,\zeta^2\,p_4^4-450\,k\,r^2\,\zeta^2\,p_4^4+5625\,k^2\,r^4\,\zeta^2\,p_4^4+r^6\left(360-1560\,k\,p_4^2+1600\,k^2\,p_4^4\right)+r^8\left(-1260\,k+4560\,k^2\,p_4^2-3200\,k^3\,p_4^4\right)+r^{10}\left(900\,k^2-3000\,k^3\,p_4^2+1600\,k^4\,p_4^4\right)+r^3\left(117\,\zeta\,p_4^2-240\,k\,\zeta\,p_4^4\right)+r^5\left(-3150\,k\,\zeta\,p_4^2+6240\,k^2\,\zeta\,p_4^4\right)+r^7\left(5625\,k^2\,\zeta\,p_4^2-6000\,k^3\,\zeta\,p_4^4\right)=0$

R_{44}: $180\,k\,\zeta^2\,p_4^4+r^2\left(-72\,p_4^2+180\,k\,p_4^4\right)+r^4\left(540\,k\,p_4^2-1200\,k^2\,p_4^4\right)+r^6\left(180\,k-2580\,k^2\,p_4^2+3800\,k^3\,p_4^4\right)+r^8\left(4500\,k^3\,p_4^2-6000\,k^4\,p_4^4\right)+r^{10}\left(-4500\,k^4\,p_4^2+4500\,k^5\,p_4^4\right)+r\left(-81\,\zeta\,p_4^2+360\,k\,\zeta\,p_4^4\right)+r^3\left(450\,k\,\zeta\,p_4^2-1200\,k^2\,\zeta\,p_4^4\right)+r^5\left(-2025\,k^2\,\zeta\,p_4^2+1800\,k^3\,\zeta\,p_4^4\right)=0$

R_{66}: $-180\,k\,\zeta^2\,p_4^4+r^2\left(72\,p_4^2-180\,k\,p_4^4\right)+r^4\left(-540\,k\,p_4^2+1200\,k^2\,p_4^4\right)+r^6\left(-180\,k+2580\,k^2\,p_4^2-3800\,k^3\,p_4^4\right)+r^8\left(-4500\,k^3\,p_4^2+6000\,k^4\,p_4^4\right)+r^{10}\left(4500\,k^4\,p_4^2-4500\,k^5\,p_4^4\right)+r\left(81\,\zeta\,p_4^2-360\,k\,\zeta\,p_4^4\right)+r^3\left(-450\,k\,\zeta\,p_4^2+1200\,k^2\,\zeta\,p_4^4\right)+r^5\left(2025\,k^2\,\zeta\,p_4^2-1800\,k^3\,\zeta\,p_4^4\right)=0$

The constant term $9\,\zeta^2\,p_4^4$ of R_{11} must be zero.

Setting the product $\zeta\,p_4$ to zero further simplifies R_{11}:

R_{11}: $r^6\left(360-1560\,k\,p_4^2+1600\,k^2\,p_4^4\right)+r^8\left(-1260\,k+4560\,k^2\,p_4^2-3200\,k^3\,p_4^4\right)+r^{10}\left(900\,k^2-3000\,k^3\,p_4^2+1600\,k^4\,p_4^4\right)=0$

The only common solution that makes each of three terms equal zero is

$$k=\frac{3}{8\,p_4^2}$$

$$\Rightarrow\quad p_4\neq 0\quad\Rightarrow\quad \zeta=0$$

$$\Rightarrow\quad R_{44}:\ \frac{135\,r^4}{4}-\frac{455\,625\,r^{10}}{8192\,p_4^6}+\frac{30\,375\,r^8}{256\,p_4^4}-\frac{6075\,r^6}{64\,p_4^2}-\frac{9}{2}\,r^2\,p_4^2=0\ \text{is false.}$$

We conclude that this assumption does not lead to a solution.

Assumption 2: $\mathcal{D}_6=\frac{1}{B}$ (inverse for the circle)

Again ignoring R_{00}, R_{33}, and R_{55}, the remaining Lambda field equations are:

R_{11}: $-\frac{5k}{a^2}+\Lambda+\frac{B'}{a^2 r B^2}-\frac{(B')^2}{a^2 B^3}+\frac{B' \mathcal{D}_4'}{2a^2 B^2 \mathcal{D}_4}+\frac{(\mathcal{D}_4')^2}{2a^2 B \mathcal{D}_4^2}+\frac{B''}{2a^2 B^2}-\frac{\mathcal{D}_4''}{a^2 B \mathcal{D}_4}=\Lambda$

R_{22}: $-\frac{5k}{a^2}+\frac{1}{a^2 r^2}-\frac{1}{a^2 r^2 B}+\Lambda+\frac{B'}{a^2 r B^2}-\frac{\mathcal{D}_4'}{a^2 r B \mathcal{D}_4}=\Lambda$

R_{44}: $-\frac{5k}{a^2}+\Lambda+\frac{1}{a^2 p_4^2 \mathcal{D}_4}-\frac{\mathcal{D}_4'}{a^2 r B \mathcal{D}_4}+\frac{B' \mathcal{D}_4'}{2a^2 B^2 \mathcal{D}_4}-\frac{\mathcal{D}_4''}{2a^2 B \mathcal{D}_4}=\Lambda$

R_{66}: $-\frac{5k}{a^2}+\Lambda+\frac{B'}{a^2 r B^2}-\frac{(B')^2}{a^2 B^3}+\frac{B' \mathcal{D}_4'}{2a^2 B^2 \mathcal{D}_4}+\frac{B''}{2a^2 B^2}=\Lambda$

R_{22} can again be solved, this time for either B or $\mathcal{D}_4$, with the latter being a simpler expression:

$\mathcal{D}_4(r)=e^{\int_1^r \frac{-B+B^2-5kB^2\rho^2+\rho B'}{B\rho}\,d\rho}$ where a constant of integration has been rolled into p_4.

This again leaves three equations in the single unknown, B:

R_{11}:
$-\frac{5k}{a^2}+\frac{2}{a^2 r^2}-\frac{3}{2a^2 r^2 B}+\frac{5kB}{a^2}-\frac{B}{2a^2 r^2}-\frac{25k^2 r^2 B}{2a^2}+\Lambda+\frac{3B'}{2a^2 r B^2}-\frac{3B'}{2a^2 r B}+\frac{15krB'}{2a^2 B}-\frac{B''}{2a^2 B^2}=\Lambda$

R_{44}: $-\frac{5k}{2a^2}+\frac{1}{2a^2 r^2}+\frac{5kB}{a^2}-\frac{B}{2a^2 r^2}-\frac{25k^2 r^2 B}{2a^2}+\Lambda+\frac{e^{-\int_1^r\left(-\frac{1}{\rho}+B\left(\frac{1}{\rho}-5k\rho\right)+\frac{B'}{B}\right)d\rho}}{a^2 p_4^2}-\frac{B'}{2a^2 r B^2}-\frac{B'}{a^2 r B}+\frac{5krB'}{a^2 B}+\frac{(B')^2}{2a^2 B^3}-\frac{B''}{2a^2 B^2}=\Lambda$

R_{66}: $-\frac{5k}{a^2}+\Lambda+\frac{B'}{2a^2 r B^2}+\frac{B'}{2a^2 r B}-\frac{5krB'}{2a^2 B}-\frac{(B')^2}{2a^2 B^3}+\frac{B''}{2a^2 B^2}=\Lambda$

$R_{11}+R_{66}$ has two distinct solutions for B:

$R_{11}+R_{66}$:
$-\frac{10k}{a^2}+\frac{2}{a^2 r^2}-\frac{3}{2a^2 r^2 B}+\frac{5kB}{a^2}-\frac{B}{2a^2 r^2}-\frac{25k^2 r^2 B}{2a^2}+2\Lambda+\frac{2B'}{a^2 r B^2}-\frac{B'}{a^2 r B}+\frac{5krB'}{a^2 B}-\frac{(B')^2}{2a^2 B^3}=2\Lambda$

The first solution to $R_{11}+R_{66}$:

$B(r)=\frac{1}{1-\frac{5kr^2}{3}+\frac{\chi}{r}}$ where χ is a constant of integration

$\Rightarrow \quad \mathcal{D}_4(r)=1$

and $\mathcal{D}_6(r)=1-\frac{5kr^2}{3}+\frac{\chi}{r}$.

This solves two of the remaining three equations:

R_{11}: $\Lambda=\Lambda$

R_{44}: $\Lambda + \frac{-5\,k + \frac{1}{p_4^2}}{a^2} = \Lambda$

R_{66}: $\Lambda = \Lambda$

R_{44} is solved when

$$k = \frac{1}{5\,p_4^2} > 0,$$

or $p_4 = \sqrt{\frac{1}{5\,k}}$.

Observe that curvature, k, is tied to the 2-sphere radius, p_4, and not to the circle radius, p_6. This is because the circle is flat but the 2-sphere has curvature $\frac{1}{{p_4}^2}$.

All of the equations are satisfied and the first 7D_2 GRW solution is

$$A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2} = c^2 \quad \text{where } k = \frac{1}{5\,p_4^2} > 0$$

$$B(r) = \frac{1}{\mathcal{D}_6[r]},$$

$$C(r) = 1,$$

$$\mathcal{D}_4(r) = 1,$$

$$\mathcal{D}_6(r) = 1 - \frac{5\,k\,r^2}{3} + \frac{\chi}{r} \quad \text{where } \chi \text{ is a free constant,}$$

and line element

$$ds^2 = \frac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2}{1 - \frac{5\,k\,r^2}{3} + \frac{\chi}{r}}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2 - a[t]^2\,p_4^2\,d\psi_4^2$$
$$- a[t]^2\,\text{Sin}[\psi_4]^2\,p_4^2\,d\psi_5^2$$
$$- \left(1 - \frac{5\,k\,r^2}{3} + \frac{\chi}{r}\right) a[t]^2\,p_6^2\,d\psi_6^2$$

Observations:

- The 2-sphere parameter, $\mathcal{D}_4$, is degenerate (i.e., $\mathcal{D}_4$=1) while the circle parameter, $\mathcal{D}_6$, is inversely proportional to B and is not degenerate.
- T^2 tolerates $\mathcal{D}_4 = \mathcal{D}_5$ (trivially); $S^2 \times S^1$ forbids $\mathcal{D}_4 = \mathcal{D}_6$ (structurally). This is proven in the Appendix.
- Because $k = \frac{1}{5\,{p_4}^2}$ and B(r) depends on k, the size of the base 3-space is limited by p_4. **This solution cannot represent our universe.**

The second solution to R_{11}+R_{66}:

$$B(r) = \frac{1}{\frac{1}{3} - k\,r^2 + \frac{\chi}{r^3}} \quad \text{where } \chi \text{ is a constant of integration}$$

$$\Rightarrow \quad \mathcal{D}_4(r) = r^2$$

$$\text{and} \quad \mathcal{D}_6(r) = \frac{1}{3} - k\,r^2 + \frac{\chi}{r^3}.$$

This again solves two of the remaining three equations:

R_{11}: $\Lambda = \Lambda$

R_{44}: $\Lambda + \frac{-1 + \frac{1}{p_4^2}}{a^2\,r^2} = \Lambda$

R_{66}: $\Lambda = \Lambda$

R_{44} is solved when

$$p_4 = 1$$

All of the equations are satisfied and we have found a second 7D_2 GRW solution:

$$p_4 = 1,$$

$$A(t,r) = \mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2} = c^2 \quad \text{where curvature k is a constant parameter,}$$

$$B(r) = \frac{1}{\mathcal{D}_6[r]},$$

$$C(r) = 1,$$

$$\mathcal{D}_4(r) = r^2,$$

$$\mathcal{D}_6(r) = \frac{1}{3} - k\,r^2 + \frac{\chi}{r^3} \quad \text{where } \chi \text{ is a free constant,}$$

and line element

$$ds^2 = \frac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2}{\frac{1}{3} - k\,r^2 + \frac{\chi}{r^3}}\,dr^2 - r^2\,a[t]^2\,d\Theta^2 - r^2\,a[t]^2\,\text{Sin}[\Theta]^2\,d\phi^2 - r^2\,a[t]^2\,d\psi_4^2$$
$$- r^2\,a[t]^2\,\text{Sin}[\psi_4]^2\,d\psi_5^2$$
$$- \left(\frac{1}{3} - k\,r^2 + \frac{\chi}{r^3}\right) a[t]^2\,p_6^2\,d\psi_6^2$$

Observations:

- $\mathcal{A}$, '$B=\frac{1}{\mathcal{D}_6}$', and C are the same in both solutions; only $\mathcal{D}_4$ and $\mathcal{D}_6$ differ.
- $\mathcal{D}_4$ cannot equal $\mathcal{D}_6$, which is consistent with the Appendix.

- Similar to the base 3-space, S^2 has $a[t]^2 r^2$ in its line element term, so its radius depends on r, which grows and thus is not a small curled dimension.
- This shows that **this solution does not represent our universe**.

b: Solve $\mathcal{R}(t) = c^2$ to find $a(t)$,

$a(t)$ is shown below. Steps for finding $a(t)$ by solving $\frac{6\,\dot{a}[t]^2}{-6k + \Lambda\, a[t]^2} = c^2$ were provided in Case 4D_a.

$$a(t) = \begin{cases} \sqrt{6}\,\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\cosh\left(\frac{c\sqrt{\Lambda}\,t}{\sqrt{6}}\right) & \text{if } k > 0 \;\&\; \Lambda \neq 0 \\ \sqrt{6}\,\sqrt{\frac{1}{\Lambda}}\,\sqrt{|k|}\,\sinh\left(\frac{c\sqrt{\Lambda}\,t}{\sqrt{6}}\right) & \text{if } k < 0 \;\&\; \Lambda \neq 0 \\ c\,t\,\sqrt{|k|} & \text{if } k < 0 \;\&\; \Lambda = 0 \\ \alpha\, e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{6}}} & \text{if } k = 0 \;\&\; \Lambda \geq 0 \;\&\; \alpha > 0 \text{ is a free constant} \end{cases}$$

Solution 1 only contains the $k > 0$ branch.

c: Kaluza-Klein (KK) bubble-of-nothing

The KK bubble has a smoothness condition that constrains parameter χ in both solutions, making it dependent on p_6 and k rather than a free parameter.
The smoothness analysis is outlined below.
See 6D_1 for a detailed explanation.

- Both solutions have a circle direction $(\mathcal{D}_6)$ that shrinks to zero size at the pinch r_0:

(1) $\mathcal{D}_6(r_0) = 0$.

✓ This is the "bubble-of-nothing" degeneracy.

Solution 1: $\mathcal{D}_6[r_0] = 1 + \frac{\chi}{r_0} - \frac{5\,k\,r_0^2}{3} = 0$

(2) $\Rightarrow \quad \chi = \frac{1}{3}\,r_0\left(-3 + 5\,k\,r_0^2\right)$

Solution 2: $\mathcal{D}_6[r_0] = \frac{1}{3} + \frac{\chi}{r_0^3} - k\,r_0^2 = 0$

(3) $\Rightarrow \quad \chi = \frac{1}{3}\,r_0^3\left(-1 + 3\,k\,r_0^2\right)$

- Substituting (2) or (3) into $\mathcal{D}_6'(r)$ gives the same result for both solutions:

$$(4)\quad \mathbb{D}_6'(r_0) = \frac{1 - 5\,k\,r_0^2}{r_0}$$

- The smoothness condition

  $$p_6\ |\mathbb{D}_6'(r_0)| = 2$$

  yields a quadratic in r_0:

  $$5\,k\,p_6\,r_0^2 \pm 2\,r_0 - p_6 = 0.$$

$$\Rightarrow \quad r_0 = \frac{\mp 1 + \sqrt{1 + 5\,k\,p_6^2}}{5\,k\,p_6}$$

- With this formula for r_0, equations (2) and (3) become closed-form formulas for χ in terms of k and p_6.
- For Solution 1, $k = \frac{1}{5\,p_4^2} > 0$, so r_0 is always real.
- For Solution 2, k is free, and r_0 is real only when $1 + 5kp_6^2 \geq 0$; i.e.,
  - ✓ $k \geq -\frac{1}{5\,p_6^2}$
  - ✓ For $k = 0$, take the limit $r_0 = \frac{p_6}{2}$, which gives $\chi = -\frac{p_6^3}{24}$.

Formula (4) is the slope of $\mathbb{D}_6$ at r_0, one step of the 6D_1 tangent-matching/periodicity condition that simplifies here to a quadratic equation in r_0.

d: Free-gauge GRW

GRW

Recall that this case has two GRW solutions.

The solutions include the same $\mathcal{A}$, B, and C:

$$\mathcal{A}(t) = c^2,$$

$$B(r) = \frac{1}{\mathbb{D}_6[r]},$$

$$C(r) = 1,$$

Solution 1 adds

$$k = \frac{1}{5\,p_4^2} > 0$$

$$\mathbb{D}_4(r) = 1,$$

$$\mathbb{D}_6(r) = 1 - \frac{5\,k\,r^2}{3} + \frac{\chi}{r}$$ where χ is a free constant.

Solution 2 adds

$p_4 = 1$,

k is a free constant,

$\mathcal{D}_4(r) = r^2$,

$\mathcal{D}_6(r) = \frac{1}{3} - k\,r^2 + \frac{\chi}{r^3}$ where χ is a free constant.

FG-GRW

We extend both GRW solutions to FG-GRW by applying the standard pattern:

- $\mathcal{A}$ becomes $\frac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2}$
- $B \to \frac{(2\,C + r\,C')^2}{4\,C\,(FG - GRW\,\mathcal{D}_6)}$
- C becomes a free parameter.
- Parameters $\mathcal{D}_4$ and $\mathcal{D}_6$ use the pattern $r^2 \to r^2 C$

Solution 1

$k = \frac{1}{5\,p_4^2} > 0$

- $\mathcal{A}(t) = \frac{6\,\dot{a}[t]^2}{\Lambda\,a[t]^2 - \frac{6}{5\,p_4^2}}$
- $B(r) = \frac{(2\,C + r\,C')^2}{4\,C\,(FG - GRW\,\mathcal{D}_6)} = \frac{(2\,C[r] + r\,C'[r])^2}{4\,C[r]\left(1 + \frac{\chi}{r\sqrt{C[r]}} - \frac{r^2\,C[r]}{3\,p_4^2}\right)}$
- $C(r)$ is a free parameter
- $\mathcal{D}_4(r) = 1$ (nothing to replace)
- $\mathcal{D}_6(r) = 1 + \frac{\chi}{r\sqrt{C[r]}} - \frac{r^2\,C[r]}{3\,p_4^2}$ (replaced r by $r\sqrt{C}$ & r^2 by $r^2 C$)

We have defined all of the parameters, so
we check whether the field equations are all identities:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{02}: $0 = 0$

R_{03}: $0 = 0$

R_{04}: $0 = 0$

R_{05}: $0 = 0$

R_{06}: $0 = 0$

R_{11}: $\Lambda = \Lambda$

R_{12}: $0 = 0$

R_{13}: $0 = 0$

R_{14}: $0 = 0$

R_{15}: $0 = 0$

R_{16}: $0 = 0$

R_{22}: $\Lambda = \Lambda$

R_{23}: $0 = 0$

R_{24}: $0 = 0$

R_{25}: $0 = 0$

R_{26}: $0 = 0$

R_{33}: $\Lambda = \Lambda$

R_{34}: $0 = 0$

R_{35}: $0 = 0$

R_{36}: $0 = 0$

R_{44}: $\Lambda = \Lambda$

R_{45}: $0 = 0$

R_{46}: $0 = 0$

R_{55}: $\Lambda = \Lambda$

R_{56}: $0 = 0$

R_{66}: $\Lambda = \Lambda$

We have found a Free-gauge GRW solution.

Solution 2

- $p_4 = 1$
- $\mathcal{R}(t) = \dfrac{6\,\dot{a}[t]^2}{-6\,k + \Lambda\,a[t]^2}$
- $B(r) = \dfrac{(2\,C + r\,C')^2}{4\,C\,(\text{Free} - \text{gauge GRW}\,\mathcal{D}_6)} = \dfrac{(2\,C[r] + r\,C'[r])^2}{4\,C[r]\left(\frac{1}{3} + \frac{\chi}{r^3\,C[r]^{3/2}} - k\,r^2\,C[r]\right)}$
- $C(r)$ is a free parameter
- $\mathcal{D}_4(r) = r^2\,C[r]$ (replaced r^2 by $r^2 C(r)$)
- $\mathcal{D}_6(r) = \dfrac{1}{3} + \dfrac{\chi}{r^3\,C[r]^{3/2}} - k\,r^2\,C[r]$ (replaced r by $r\sqrt{C[r]}$ & r^2 by $r^2 C[r]$

With these formulas, all of the equations are satisfied:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{02}: 0 = 0

R_{03}: 0 = 0

R_{04}: 0 = 0

R_{05}: 0 = 0

R_{06}: 0 = 0

R_{11}: Λ = Λ

R_{12}: 0 = 0

R_{13}: 0 = 0

R_{14}: 0 = 0

R_{15}: 0 = 0

R_{16}: 0 = 0

R_{22}: Λ = Λ

R_{23}: 0 = 0

R_{24}: 0 = 0

R_{25}: 0 = 0

R_{26}: 0 = 0

R_{33}: Λ = Λ

R_{34}: 0 = 0

R_{35}: 0 = 0

R_{36}: 0 = 0

R_{44}: Λ = Λ

R_{45}: 0 = 0

R_{46}: 0 = 0

R_{55}: Λ = Λ

R_{56}: 0 = 0

R_{66}: Λ = Λ

We have found a second Free-gauge GRW solution.

e: The Vacuum Friedmann Equations

We only need to display the Friedmann equations once.

The equations involve only k, so both solutions generate the same equations

Plugging the GRW Solution 2 formulas for $\mathcal{A}$, B, C, D_4, and D_6 (but not a)

into the field equations yields:

R_{00}: $\frac{6\,\ddot{a}}{a\,c^2} = \Lambda$

R_{11}: $\frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22}: $\frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{33}: $\frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{44}: $\frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{55}: $\frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{66}: $\frac{5\,k}{a^2} + \frac{5\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$

R_{22}-R_{66} are redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$-\frac{1}{30}R_{00}+\frac{1}{5}R_{11}$: $\frac{k}{a^2} + \frac{\dot{a}^2}{a^2\,c^2} = \frac{\Lambda}{6}$

$\frac{1}{6}R_{00}$: $\frac{\ddot{a}}{a\,c^2} = \frac{\Lambda}{6}$

Appendix: $\mathbb{D}_4$ cannot equal $\mathbb{D}_6$

Assume $\mathbb{D}_4 = \mathbb{D}_6$; we derive a contradiction.

We don't assume any relationship to $\mathbb{B}$.

The initial Lambda field equations simplify to:

R_{11}: $-\frac{5\,k}{a^2} + \Lambda + \frac{\mathbb{B}'}{a^2\,r\,\mathbb{B}^2} + \frac{3\,\mathbb{B}'\,\mathbb{D}_4'}{4\,a^2\,\mathbb{B}^2\,\mathbb{D}_4} + \frac{3\,(\mathbb{D}_4')^2}{4\,a^2\,\mathbb{B}\,\mathbb{D}_4^2} - \frac{3\,\mathbb{D}_4''}{2\,a^2\,\mathbb{B}\,\mathbb{D}_4} = \Lambda$

R_{22}: $-\frac{5\,k}{a^2} + \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,\mathbb{B}} + \Lambda + \frac{\mathbb{B}'}{2\,a^2\,r\,\mathbb{B}^2} - \frac{3\,\mathbb{D}_4'}{2\,a^2\,r\,\mathbb{B}\,\mathbb{D}_4} = \Lambda$

R_{44}: $-\frac{5\,k}{a^2} + \Lambda + \frac{1}{a^2\,p_4^2\,\mathbb{D}_4} - \frac{\mathbb{D}_4'}{a^2\,r\,\mathbb{B}\,\mathbb{D}_4} + \frac{\mathbb{B}'\,\mathbb{D}_4'}{4\,a^2\,\mathbb{B}^2\,\mathbb{D}_4} - \frac{(\mathbb{D}_4')^2}{4\,a^2\,\mathbb{B}\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{2\,a^2\,\mathbb{B}\,\mathbb{D}_4} = \Lambda$

R_{66}: $-\frac{5\,k}{a^2} + \Lambda - \frac{\mathbb{D}_4'}{a^2\,r\,\mathbb{B}\,\mathbb{D}_4} + \frac{\mathbb{B}'\,\mathbb{D}_4'}{4\,a^2\,\mathbb{B}^2\,\mathbb{D}_4} - \frac{(\mathbb{D}_4')^2}{4\,a^2\,\mathbb{B}\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4''}{2\,a^2\,\mathbb{B}\,\mathbb{D}_4} = \Lambda$

R_{44}-R_{66}gives $\frac{1}{a^2\,p_4^2\,\mathbb{D}_4}=0$, which is impossible:

$$R_{44}-R_{66}: \quad \frac{1}{a^2\, p_4^2\, \mathbb{D}_4} = 0$$

This result has an interesting implication:

- Case 6D_1: $T^2 = S^1 \times S^1$ tolerates $\mathbb{D}_4 = \mathbb{D}_5$ (trivially):
  - ✓ Because both factors are equally flat, no obstruction exists.
- Case 7D_2: $S^2 \times S^1$ forbids $\mathbb{D}_4 = \mathbb{D}_6$ (structurally)
  - ✓ This is a 'curved S^2 vs flat S^1' obstruction,
  - ✓ No relative scaling between the S^2 and S^1 factors can fix this